

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Data Warehousing and Data Mining

COURSE CODE:CSA1603

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

<i>Criteria</i>	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I P. GAYATHRI/192425277 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.GAYATHRI

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

- 1 600
- 2 350
- 3 220
- 4 200
- 5 190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

	User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
User Data	U1	120	18	2000
	U2	60	5	5000
	U3	140	22	1500
	U4	50	4	6000
	U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

	Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85	
F2	500	4	70	
F3	400	6	80	
F4	280	9	90	
F5	520	3	65	

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score			
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

CSA1603 - Data warehousing and mining

CO4 - ATL 1

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Reg No: 192425277

① Dataset: Hospital patient Segmentation

Patient ID	Age	monthly medical Expenses(₹)	Health Risk score
P ₁	28	12000	65
P ₂	55	25000	40
P ₃	42	20000	75
P ₄	30	11000	70
P ₅	60	28000	85

SSE values:-

K	SSE
1	600
2	350
3	220
4	200
5	190

Q. A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the following elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

SSE - Means $\rightarrow$ sum of squared errors

we are using elbow method

vivo T3x

Aug 21, 2026, 08:23

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -i 500 -num-slots 1 -S 10

Cluster mode:
☒ Use training set
☐ Supplied test set
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) HealthRisk
☒ Store clusters for visualization

Ignore attributes:

Start Stop

Result list (right-click for options):
 11:41:37 - SimpleKMeans
 11:44:04 - SimpleKMeans

Status: OK

Cluster output:

```

=== Run information ===

Scheme:      weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -
Relation:     hospital
Instances:    5
Attributes:   3
  Age
  MedicalExpense
  HealthRisk

Test mode:    evaluate on training data

=== Clustering model (full training set) ===

KMeans
=====

Number of iterations: 3
Within cluster sum of squared errors: 0.047086194312283745

Initial starting points (random):

Cluster 0: 30,11000,70
Cluster 1: 55,25000,40
Cluster 2: 28,12000,45

Missing values globally replaced with mean/mode
  
```

Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -i 500 -num-slots 1 -S 10

Cluster mode:
☒ Use training set
☐ Supplied test set
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) HealthRisk
☒ Store clusters for visualization

Ignore attributes:

Start Stop

Result list (right-click for options):
 11:41:37 - SimpleKMeans
 11:44:04 - SimpleKMeans

Status: OK

Cluster output:

```

Cluster 0: 30,11000,70
Cluster 1: 55,25000,40
Cluster 2: 28,12000,45

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute      Full Data      Cluster0      Cluster1      Cluster2
-----
Age            43             30.0          55.0          28.0
MedicalExpense 19200          11000.0       25000.0       12000.0
HealthRisk     57             70.0          40.0          45.0

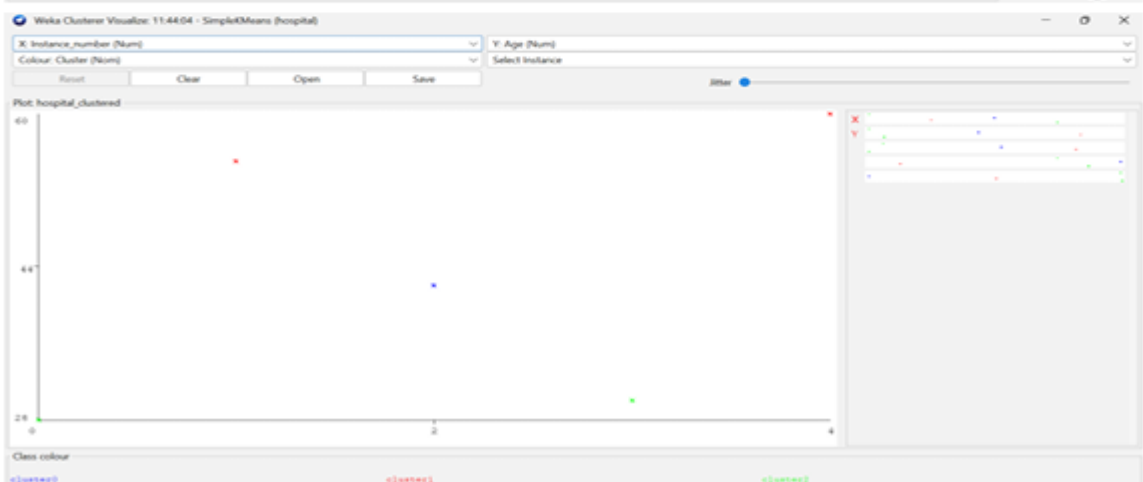
Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0  1 ( 20%)
1  2 ( 40%)
2  2 ( 40%)
  
```

Log



K	SSE	Reduction From previous k
1	600	—
2	350	250
3	220	130
4	200	20
5	190	10

We observe, that when

$k=2$ & 3 the SSE value is been reduced much and when k value is 4 & 5 the decrease is very small. This variation happens after the value of k is 3 . Therefore Elbow method suggest the value of k is 3 .

∴ Based on health risk score 3 clusters have formed

- i) Low risk - with low expense
- ii) moderate risk - with moderate expense
- iii) High risk - with high expense

Impact on Decision making:

- i) Preventive care
- ii) Routine monitoring

- ② A retailer applies Hierarchical Agglomerative clustering [HAC] to customer orders using using appropriately encoded Product category information and standardized order value. A dendrogram is generated. Interpret how clusters are progressively numbered, the clusters by selecting a suitable cut-off level on the dendrogram. Explain how the selected cluster can help understand customer purchasing behaviour.

Order ID	product category	order value (£)
1	electronics	55000
2	clothing	25000
3	electronics	57000
4	furniture	30000
5	clothing	27000

HAC

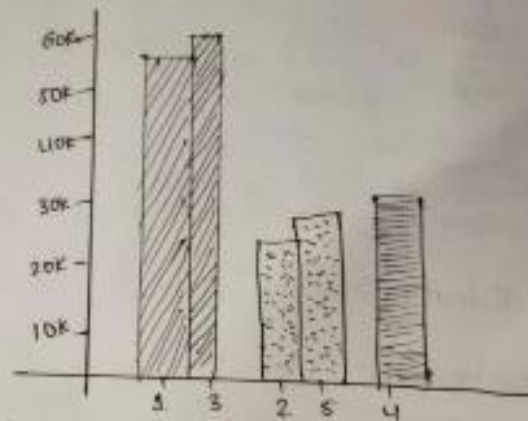
→ It calculates the distance between the clusters

→ merges the two closest clusters

→ 01 & 03 forms 1st cluster - electronics

→ 02 & 05 forms 2nd cluster - clothing

→ 04 forms 3rd cluster - furniture



But the optimal no. of clusters can not be determined by the given data.

step

1) one-hot encoding

Order ID	electronics	clothing	furniture
1	1	0	0
2	0	1	0
3	1	0	0
	0	0	1
	0	1	0

step 2: Normalize order value

• We use min-max normalization

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

• Here $x_{\min} = 25,000$

$x_{\max} = 57,000$

• Therefore

$$01. \frac{55000 - 25000}{57000 - 25000} \Rightarrow \frac{30000}{32000} \Rightarrow 0.9375$$

$$02. \frac{25000 - 25000}{57000 - 25000} \Rightarrow \frac{0}{32000} \Rightarrow 0$$

$$03. \frac{57000 - 25000}{57000 - 25000} \Rightarrow \frac{32000}{32000} \Rightarrow 1$$

$$04. \frac{30000 - 25000}{57000 - 25000} \Rightarrow \frac{5000}{32000} \Rightarrow 0.15625$$

$$05. \frac{27000 - 25000}{57000 - 25000} \Rightarrow \frac{2000}{32000} \Rightarrow 0.0625$$

step 3: Calculate Euclidean distances.

For two observations:

$$d(P, Q) = \sqrt{\sum (P_i - Q_i)^2}$$

1. E13

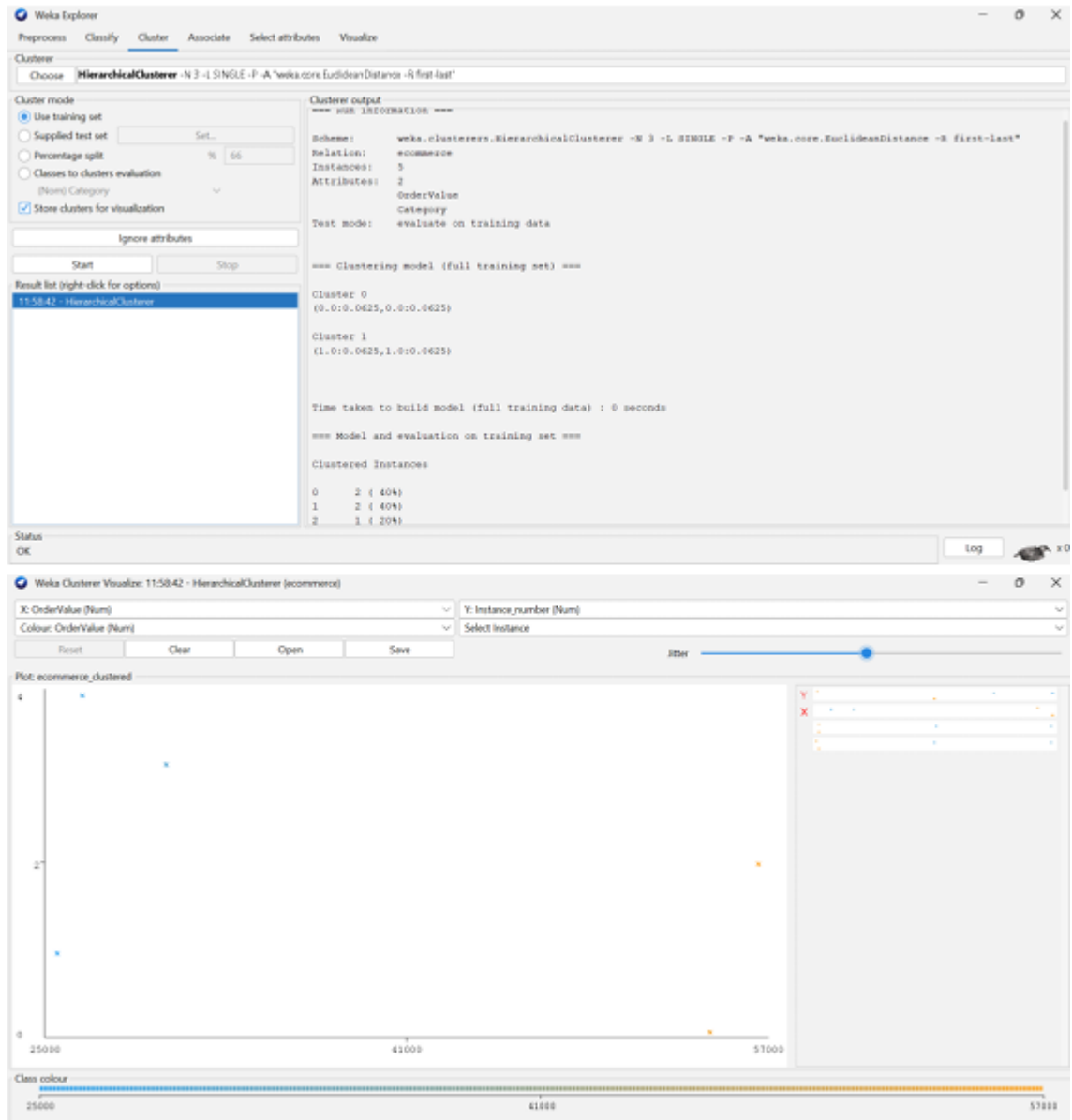
$$\begin{aligned} d(01, 02) &= \sqrt{(0-1)^2 + (0-0)^2 + (0-0)^2 + (0.9375-1)^2} \\ &= \sqrt{0+0+0+(-0.0625)^2} \\ &= \sqrt{0.00390625} = 0.0625 \end{aligned}$$

2. E15

$$d(2, 5) = |0 - 0.0625|$$

$$= 0.0625$$

$$d(04) = 0.0625$$



Step 4

ID	1	2	3	4	5
1	0	1.697	0.063	1.616	1.663
2	1.697	0	1.732	1.423	0.063
3	0.063	1.732	0	1.647	1.697
4	1.616	1.423	1.647	0	1.417
5	1.663	0.063	1.697	1.417	0

Steps: Hierarchical Agglomerative clustering

$$D(A, B) = \min d(x, y)$$

between members of the two clusters merged

Smallest distance:

$$1-3 \Rightarrow 0.0625$$

By seeing smallest distance

$$C_1 = 01 + 03 \quad ; \quad C_1 = \{01, 03\} \Rightarrow 0.0625$$

$$C_2 = 02 + 05 \quad ; \quad C_2 = \{02, 05\} \Rightarrow 0.0625$$

$$C_3 = 04 \quad ; \quad C_3 = \{04\} \Rightarrow 0.0625$$

Merging the clusters can be done by finding the distance between C_1 & C_3 ; C_2 & C_3 minimum of $C_1 \& C_3$, $C_2 \& C_3$

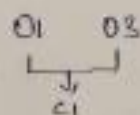
$$C_1 \& C_3$$

$$(01, 03) \& C_3$$

$$\min(01, 04), (03, 04)$$

$$\min(1.616, 1.647)$$

$$1.616$$



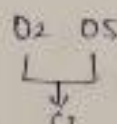
$$C_2 \& C_3$$

$$(02, 05) \& C_3$$

$$\min(02, 04), (05, 04)$$

$$\min(1.452, 1.417)$$

$$1.417$$



3) Dataset: mobile app user Behaviour

Usage	Daily Usage	Logic frequency	In app purchase
120	18	2000	

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U ₂	60	5	5000
U ₃	110	22	1500
U ₄	50	4	6000
U ₅	130	20	1800

Therefore:

Step 1: Identify the features

We have 3 features for each user:

- * Usage time (minutes)
- * Login frequency
- * purchase amount (₹)

Step 2: Observe the Data

Look at the user's behaviour

U₁, U₃, U₅: * High Usage
* High Login frequency

So they form one similar group

U₂, U₄: * Low Usage
* Low Login frequency

Step 3: Compare the two groups

Features	G ₁ : U ₁ , U ₃ , U ₅	G ₂ : U ₂ , U ₄
Usage	High	Low
Logins	High	Low
purchase	Low	High

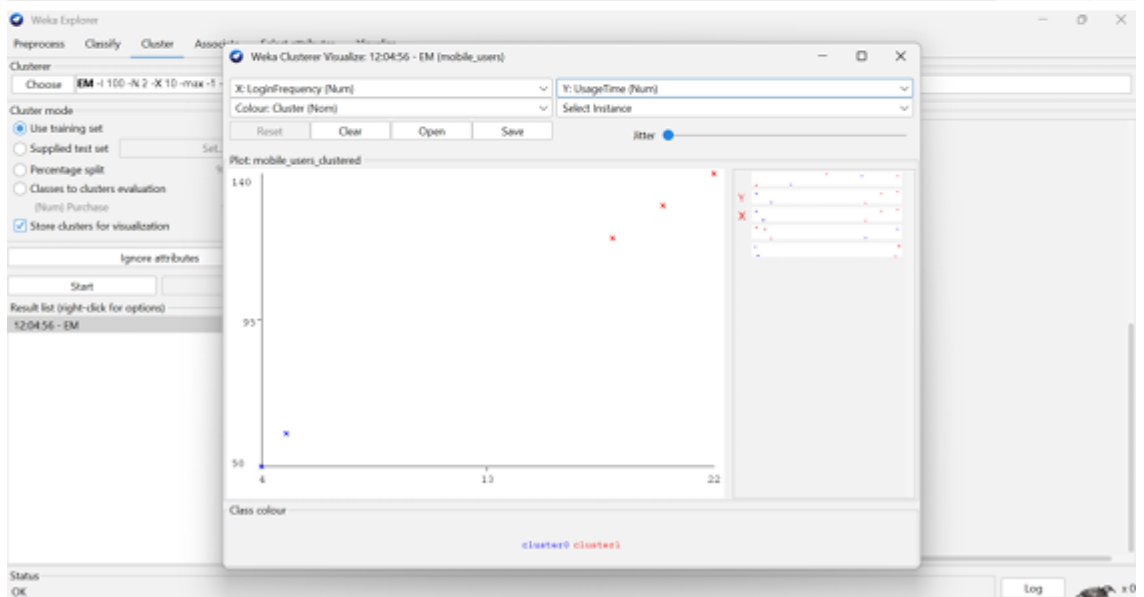
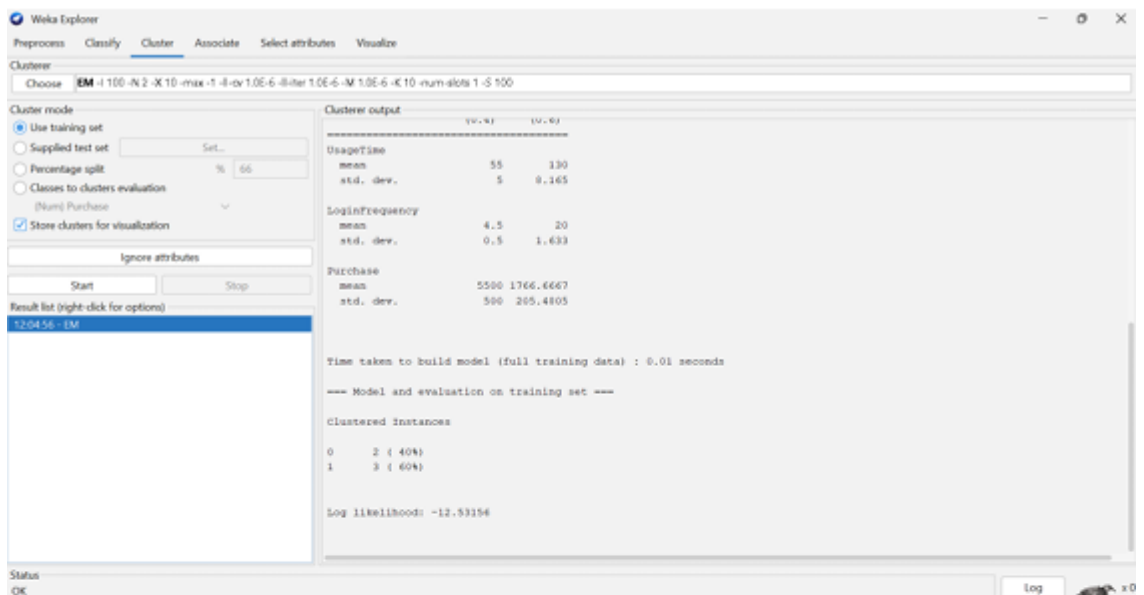
Step 4: Determine No. of clusters

Since there are two distinct behavioural pattern we choose
: K = 2.

Therefore:

Vivo T3x

No. of clusters = 2



Step 5: Name the clusters

cluster 1 - highly engaged, lower - purchase users
→ U1, U3, U5

cluster 2 - less engaged, high - purchase users
→ U2, U4

Step 6: Final Result

K=2

4) Dataset: Agricultural crop monitoring

Farm	Rainfall (mm)	soil fertility score	crop yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Step 1 calculate distances before normalization we use Euclidean distance.

$$d(P, B) = \sqrt{\sum_i (P_i - B_i)^2}$$

for example

$$\begin{aligned} d(F_1, F_4) &= \sqrt{(300-280)^2 + (8-9)^2 + (85-90)^2} \\ &= \sqrt{400+1+25} \\ &= \sqrt{426} \approx 20.64 \end{aligned}$$

similarity the important distance are:

pair	Distance
F1 - F4	20.64
F2 - F5	20.64
F1 - F3	100.14
F2 - F3	100.52
F3 - F4	120.45
F3 - F5	120.77

$$F_1 - F_2 = 200.60$$

$$F_4 - F_5 = 201.34$$

HAC before normalization

using single linkage The smallest distance is:

$$F_1 - F_4 = 20.64$$

and

$$F_2 - F_5 = 20.64$$

Therefore:

$$C_1 = \{F_1, F_4\}$$

Next, F_3 joins

$$C_2 = \{F_2, F_5\}$$

C_1 , because:

$$C_3 = \{F_3\}$$

$$\min[d(F_3, F_1), d(F_3, F_4)] = \min(100.4, 120.45) = 100.4$$

So

$$C_1 = \{F_1, F_2, F_3, F_4\}$$

Step 2:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

- Rainfall : $x_{\min} = 280$, $x_{\max} = 520$
- Soil Fertility : $x_{\min} = 3$, $x_{\max} = 9$
- Crop Yield : $x_{\min} = 65$, $x_{\max} = 90$

Step 3: Euclidean Distance

Pair	Distance
$F_1 - F_4$	0.2734
$F_1 - F_2$	0.2734
$F_2 - F_5$	0.5698
$F_1 - F_3$	0.6667
$F_2 - F_3$	0.9324
$F_3 - F_4$	0.9294
$F_3 - F_5$	

Step 4: HAC clustering

3 clusters:

$$C_1 = \{F_1, F_4\}$$

$$C_2 = \{F_2, F_5\}$$

$$C_3 = \{F_3\}$$

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster: Choose **SimpleKMeans** int 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -it 1 -N 3 -A "wks core EuclideanDistance -R first-last" -1500 -num-cls 1 -S 10

Cluster mode:

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes:

Start Stop

Result list (right-click for options): 12:15:11 - SimpleKMeans

Cluster output:

```

--- Run information ---
Scheme: wks.core.EuclideanDistance -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -it 1 -N 3 -A "wks core EuclideanDistance -R first-last" -1500 -num-cls 1 -S 10
User action: agriculture
Subset count: 5
Rainfall: 5
SoilFertility: 5
CropField: 5
Best model: evaluate on training data

--- Filtering model (full training set) ---

KMeans
=====
Number of iterations: 3
Within-cluster sum of squared errors: 6.676703033333333
Initial clustering points (random):
Classes: 0: 200, 8, 80
         1: 500, 4, 70
         2: 300, 6, 83
ML10000 ML10000 45000000 00000000 00000000
  
```

Status OK Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster: Choose **SimpleKMeans** int 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -it 1 -N 3 -A "wks core EuclideanDistance -R first-last" -1500 -num-cls 1 -S 10

Cluster mode:

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes:

Start Stop

Result list (right-click for options): 12:15:11 - SimpleKMeans

Cluster output:

```

--- Run information ---
Scheme: wks.core.EuclideanDistance -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -it 1 -N 3 -A "wks core EuclideanDistance -R first-last" -1500 -num-cls 1 -S 10
User action: agriculture
Subset count: 5
Rainfall: 5
SoilFertility: 5
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Best model: evaluate on training data

--- Filtering model (full training set) ---

KMeans
=====
Number of iterations: 3
Within-cluster sum of squared errors: 6.676703033333333
Initial clustering points (random):
Classes: 0: 200, 8, 80
         1: 500, 4, 70
         2: 300, 6, 83
ML10000 ML10000 45000000 00000000 00000000
  
```

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (5.0)	0 (2.0)	1 (2.0)	2 (1.0)
Rainfall	400	290	510	400
SoilFertility	6	8.5	3.5	6
CropField	78	87.5	67.5	80

Time taken to build model (full training data) : 0.02 seconds

--- Model and evaluation on training set ---

Clustered Instances

Class	Count	Percentage
0	2	(40%)
1	2	(40%)
2	1	(20%)

Status OK Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster: Choose **SimpleKMeans** int 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -it 1 -N 3 -A "wks core EuclideanDistance -R first-last" -1500 -num-cls 1 -S 10

Cluster mode:

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes:

Start Stop

Result list (right-click for options): 12:15:11 - SimpleKMeans

Weka Cluster Visualize: 12:15:11 - SimpleKMeans (agriculture)

X Instance number (Num) Y Rainfall (Num)

Colour: Cluster (Num) Select Instance

Reset Clear Open Save Filter

Plot: agriculture, clustered

Class colour: 0:blue 1:green 2:red

Status OK Log